DELIRIUM



Postoperative delirium is one of the most common complications after major surgery, affecting 15–25% of older adults undergoing major non-cardiac surgery.

It is frequently underrecognized, particularly the hypoactive subtype. Delirium is associated with increased complications, longer hospital stays, loss of independence, mortality, healthcare costs, and long-term cognitive and functional decline. [1-6]

Screening Tools

Identification of patients at increased risk of delirium is an important component of prevention, allowing targeted interventions to reduce postoperative delirium. The strongest risk factors include cognitive impairment, older age, and frailty. Other important risk factors include a history of delirium, visual or hearing impairment, substance dependence, social isolation, severe comorbid illness, and postoperative factors such as infection and medications (e.g., anticholinergics, opioids, and benzodiazepines). [7]

The modified Delirium Elderly At-Risk (mDEAR) tool is recommended as the preferred screening tool because it is the most evidence-supported instrument that can be completed using patient-reported information alone. It assesses cognitive impairment, age, functional dependence, sensory impairment, and chronic substance use, with a score of ≥3 indicating increased risk of postoperative delirium. [8]

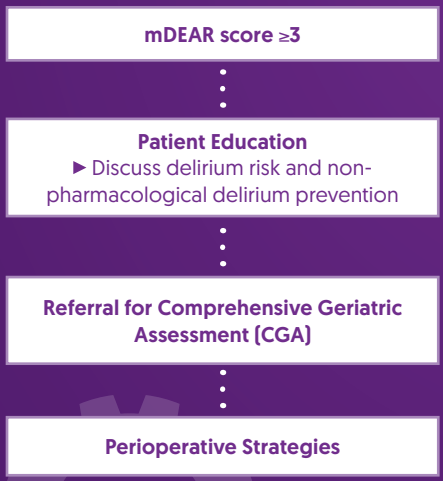
Although the evidence supporting the mDEAR remains limited and other validated delirium risk assessment tools are available, these require cognitive testing or other clinician-administered assessments that cannot be incorporated into a standardized patient-completed questionnaire. The mDEAR therefore represents the most practical option for province-wide preoperative screening while recognizing that clinical judgment should always complement screening results when assessing an individual's risk of postoperative delirium.

Modified Delirium Elderly at Risk (mDEAR) Tool

	YES	NO
1. Cognitive Impairment - Has the patient been disoriented to person, place, time or situation? - Does patient have a history of confusion or forgetfulness? - Does the patient have a history of cognitive impairment, memory loss, inattention or difficulty concentrating?	☐	☐
2. Patient Age - Age ≥ 80 years old	☐	☐
3. Sensory Impairment - Patient is hearing impaired; **OR** - Patient has low vision	☐	☐
4. Functional Status - Dependence in 1 or more activity of daily life (ADL)	☐	☐
5. Substance Use - Patient consumes > 3 drinks of alcohol per week; **OR** - Patient takes benzodiazepines > 3 times per week Examples include: Alprazolam (Xanax), Clonazepam (Klonopin), Diazepam (Valium), Lorazepam (Ativan) and Temazepam (Restoril)	☐	☐

SCORING
Assign 2 points for question 1, and 1 point each for questions 2-5

DELIRIUM ALGORITHM



Prehabilitation and Optimization Recommendations

Patient Education

- Discuss increased risk of postoperative delirium
- Review signs and symptoms
- Non-pharmacological delirium prevention:
  - Regular reorientation and cognitive stimulation
  - Access to natural light and a visible clock
  - Access to usual hearing aids and glasses
  - Early mobilization
  - Family presence, familiar items
  - Sleep hygiene (e.g. avoid daytime napping)
- Refer to online patient resources

Referral for Comprehensive Geriatric Assessment (CGA)

- Studies have shown that CGA can reduce the risk of postoperative delirium.
- Geriatrician involvement may lead to more accurate identification of delirium and proactive initiation of multimodal delirium risk management in higher risk patients. [6]
- If specialized CGA is not available in your area, consider referral to virtual provincial geriatrician resources

Perioperative Strategies

- Minimize polypharmacy and deliriogenic medications (e.g. benzodiazepines, zopiclone, anti-cholinergic medications, opioids)
- Avoid prolonged fasting of fluids
- Implement regular screening for delirium (e.g. CAM, 3D-CAM (10), 4AT)
- Oral and nutritional assistance, orienting communication and early mobilization together reduced delirium by 56% and LOS by 2 days in one study (11)
- Non-pharmacological sleep enhancement protocols (music, noise reduction, sleep aids such as melatonin)
- Optimize pain management
- Anesthetic modality has not been shown to impact incidence of delirium (12)
- Intraoperative dexmedetomidine use may prevent postoperative delirium compared to other anesthetic agents (13)
- Current evidence that depth of anesthesia monitoring and avoidance of “deep” anesthesia (i.e. BIS target 40-60) is conflicting (14), although the effect likely varies by population
- Prophylactic dosing and routine treatment with antipsychotic medications does not reduce the incidence or duration of delirium (15)